

FaceEat: Facial and Eating Activities Recognition with Inertial and Mechanomyography Fusion using a Glasses-Based Design for Real-Time and on-the-Edge Inference

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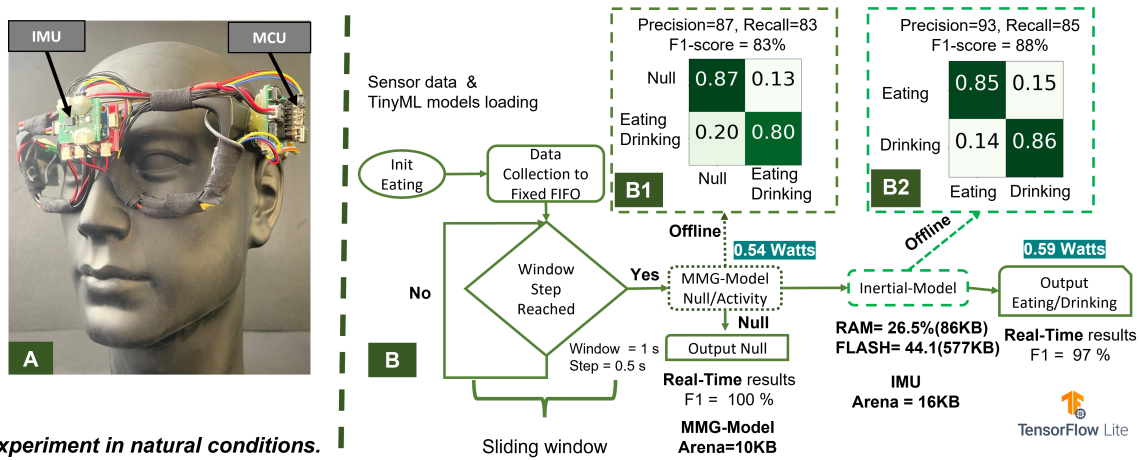


Figure 1: (A) FaceEat Prototype: On the Nose Bridge are the Inertial Measurement Unit, Microphone, Pressure and Gas Sensors. On the Temples are Pairs of Piezoelectric Film and Force Sensitive Resistor (Mechanomyography). (B) RTE Implementation for Eating/Drinking Monitoring.(B1-B2) Results for 5-Fold Cross Validation (Five Users) for Null and Eating/Drinking Recognition.

ABSTRACT

Facial expressions recognition and eating monitoring technologies can detect stress levels and emotional triggers that lead to unhealthy eating behaviors. Wearables offer a ubiquitous solution to help the individual develop coping mechanisms to manage stress and maintain a healthy lifestyle. Introducing FaceEat, a privacy-focused, real-time, and on-the-edge (RTE) wearable solution with minimal power consumption (≤ 0.55 Watts) and utilizing a tiny memory space (11 – 19KB). Its purpose is to recognize facial expressions and eating/drinking activities. At the heart of FaceEat are lightweight convolutional neural networks, serving as the backbone models for both facial and eating scenarios. During the RTE evaluation, the system achieved an F1-score of over 86% in facial expression recognition. Additionally, we achieved an F1-score of 90% for monitoring

eating and drinking activities for the user-independent case with an unseen volunteer for the RTE.

CCS CONCEPTS

• Computer systems organization → Embedded systems.

KEYWORDS

Real-Time, TinyML, Facial Expressions, Eating Monitoring, Activity Recognition

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